

Business Logic Document For External Review

Prepared on 2026-06-28 for Trade Readiness Copilot. This document explains how the product makes local preparation decisions while keeping real-world claims closed.

2-Minute Summary

The product's business logic is now implemented for the local repo-controlled scope. It can prepare a packet, compute safe statuses, score the packet, show evidence gaps, and decide which local actions are allowed.

The logic deliberately does not approve trade action. It blocks stronger claims until real source freshness, buyer/supplier evidence, qualified review, security/privacy review, payment review, and public go/no-go approval exist.

Business Position

- Product name: Trade Readiness Copilot.
- First wedge: foreign exporters preparing to sell into Canada.
- First categories: food, seafood, agri, and general goods.
- First persona: beginner-to-intermediate exporter.
- Country path: Canada destination, Vietnam demo origin, India strategic next origin, Generic fallback.
- Core object: Trade Readiness Packet.

Implemented Business Rules

- 12-question decision tree: trade direction, product, origin, destination, HS code, buyer/importer, importer of record, Incoterms, documents, regulated-product risk, official sources, and next safe move.
- Starter flow: checks minimum inputs and allows a starter packet when product, country, direction, intended use, and source routes exist.
- Market signal: computes a bounded local signal from source routes and evidence, capped before real demand proof.
- Country packs: checks whether required import, tariff, regulated-product, restricted-party, and market/buyer source routes exist.
- Source freshness: evaluates attached evidence for freshness, last verification, content hash, and review status.
- Buyer/supplier evidence: uses evidence ladders and blocks buyer-validated and supplier-verified language.
- Business gate decision: allows local drafts and reports while blocking outreach, payment, approvals, and shipment decisions.
- Phase coverage: exposes 13 business phase surfaces, while phase 0 remains the business identity lock.

Document Logic Implemented Now

Document logic is now part of the packet decision system. The product accepts no-document intake as a valid beginner path and accepts PDFs as draft evidence only when upload checks pass.

- No-document intake creates a missing-evidence packet and does not show extracted-field confirmation or delete-uploaded-files actions.
- Uploaded-document intake creates quarantine metadata, draft extraction rows, user-confirmation status, and deletion actions.
- Expected document classes covered locally: 11.
- Official sample PDFs stored for parser orientation: 3.
- Synthetic filled parser QA PDFs stored for local parser testing: 11.
- Official samples and synthetic QA files do not count as customer proof.
- Every extracted field has provenance, confidence, user-confirmation status, claim boundary, supported claims, and blocked claims.

Claim Gate Logic Implemented Now

The product now checks packet statements through a claim gate before showing them. This prevents a source route, user input, parser draft, or missing-document placeholder from being presented as real approval.

- Claim decisions generated: 17.
- Safe preparation/source-routing statements: 7.
- Blocked statements: 10.
- HS candidate, tariff route, CFIA route, market signal, and buyer lead-route statements can be shown only as preparation language.
- Origin evidence, supplier evidence, Incoterms responsibility, and document extraction stay blocked when the required proof is missing.
- Tariff confirmed, CFIA approved, buyer validated, supplier verified, customs ready, and shipment approved remain forbidden external claims.

Decision Scoring Logic Implemented Now

The product explains decisions with six separate capped scores. It does not collapse the packet into a single readiness score, because that would hide risk.

- Score records generated: 6.
- Market signal score shows whether deeper validation is worth doing, not whether demand is proven.
- Evidence completeness score shows what is missing, not whether the packet is approved.
- Source freshness score is capped when official/reference evidence is stale, reference-only, or unreviewed.
- Buyer/supplier evidence score cannot say buyer validated or supplier verified.
- Decision safety score remains red while forbidden external claims are blocked.

AI Copilot Logic Implemented Now

AI is treated as a drafting and organization helper, not as a decision maker. Deterministic rules decide what can be shown.

- AI role contracts generated: 8.
- AI output contracts generated: 8.
- Prompt-injection checks generated: 2.
- AI can produce drafts, source summaries, confirmation tasks, reviewer work orders, redaction prompts, and QA findings.
- AI cannot approve customs, tariff, CFIA, buyer, supplier, payment, shipment, legal, or launch claims.

Expert Review Logic Implemented Now

Human review is now a product workflow with reviewer lanes, credential requirements, scoped requests, finding templates, and claim-gate impact records.

- Reviewer lanes generated: 10.
- Profile requirement records: 10.
- Scoped review requests: 10.
- Gate-impact rows: 19.
- A reviewer can approve only the exact scope they reviewed, and only after credential evidence, artifacts reviewed, evidence attachments, source checks, and a dated finding are stored.
- The local product does not record completed reviews or treat review routing as approval.

Report Logic Implemented Now

Reports are now generated as first-class packet views. They organize evidence for people, but they cannot remove uncertainty or blocked claims.

- Report types generated: 12.
- Report records generated: 12.
- Exports generated: 36.
- Citation rows generated: 132.
- Starter, market, buyer-ready, supplier-request, broker-review, missing-evidence, blocked-claim, country-source, source-freshness, expert-summary, executive-decision, and audit reports are produced from the packet.
- Reports carry version, draft watermark, review status, source citations, evidence citations, and the blocked-claims section.

Current Sample Packet Result

- Packet reviewed: packet - frozen - tuna - canada - 001.

- Business logic status: `business_logic_implemented_with_external_evidence_gates`.
- Business phase surfaces: 13.
- Market signal score: 55 out of 59 before external evidence.
- Buyer evidence level: `no_evidence`.
- Supplier evidence level: `supplier_named`.
- Source freshness: `source_freshness_blocked_until_refresh_and_review`.
- Customer-visible decision: `draft_packet_allowed_external_claims_blocked`.

Actions Allowed Locally

- approve trade action: False
- generate broker or expert packet: True
- generate buyer packet draft: True
- generate missing evidence report: True
- generate starter packet: True
- refresh sources record: True
- send outreach: False
- take payment: False

Claims Still Blocked

- customs or tariff advice
- cfia or regulated product clearance
- buyer validation claim
- supplier verification or recommendation
- shipment or export ready claim
- live payment checkout
- public launch approval

Evidence Needed Before Stronger Claims

- Dated official-source refresh and qualified review.
- Qualified customs/trade review for country, tariff, CFIA, import controls, and claim language.
- Dated buyer reply, call note, LOI, PO, or paid-order evidence before buyer-demand claims.
- Supplier registration, export ability, product documents, inspection/certificate, and prior-shipment evidence before supplier confidence claims.

- Security, privacy, AI-safety, DevOps, billing/payment, report-language, and launch-owner approval before hosted/private/public rollout.

Reviewer Decision Requested

Please review whether the business rules make sense for a safe trade-readiness preparation product.

- Are any rules too weak or too strict for a controlled private-beta candidate?
- Do any labels overclaim approval, compliance, buyer validation, supplier verification, or market demand?
- Are the buyer and supplier evidence ladders reasonable?
- What must change before this can be shown to real users?